LONG NGUYEN | Technical Artist & Game Developer

Shaders · VFX · Pipelines · Artist Tools

LONG NGUYEN

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20 Web3 game maps, about 1 every 3 weeks

60+ FPS on WebGL

<300 draw calls

80%+ less map integration time

SUMMARY

Technical artist with a software engineering background and 4+ years building cross-platform games (mobile, PC, WebGL) in Unity and Unreal Engine, connecting art and engineering through real-time shaders, VFX pipelines, performance optimization, and Unity Editor tools, including AI-assisted workflows.

EXPERIENCE

Game Developer & Technical Artist – Hypereel – Yield Guild Games

Nov 2024 – Present · Part of Yield Guild Games until January 2026 · Full-time · On-site · Ho Chi Minh City

BUILDING

- Developed gameplay, UI, and **internal Unity tools** for a live WebGL sports board game.
- Main contributor to its **match mini-game**, including its skill and special card systems.
- Co-developed the **multi-board game flow** and built its **treasure services** across **10 boards**.
- Delivered gameplay and UI for **20 shipped mini-game maps** on a **Web3 social board game**, at about one map every 3 weeks.

VISUAL

- Designed **4 featured effects** per map plus a **reusable environment-effect library** shared across **all 20 maps**, using **custom shaders** (displacement, procedural patterns, math-driven effects).
- Created VFX for the sports game's **skill and special cards**.
- Kept the Web3 game's WebGL builds **above 60 FPS** and draw calls **below 300** by profiling with **Unity Profiler, Frame Debugger, and RenderDoc**, then optimizing with **LOD and texture compression**.

WORKFLOW

- Designed and implemented **automation tools** that shortened **map integration** from **about two months to about one week** across data, assets, API wiring, and gameplay logic, and reduced human error in import/export, debugging, and testing.
- Supported the 2D, 3D, and VFX teams by resolving **pipeline, in-engine, platform, and asset-to-runtime issues**.
- Worked with game designers to build **new gameplay mechanics**.

PIPELINE

- Built **Unity asset pipeline tools** for avatars and character models, with a **batch-mode import** that pushes review branches.
- Created **QA and authoring tools**: **Card Preview**, **In-Game Avatar Changer**, and **Walk Plan route authoring** in the Map Tool.
- Extended GUID asset tools with **GPU duplicate texture detection** and a **command interface** for automation.

Game Developer – Indigames Inc.

Apr 2022 – Apr 2024 · Full-time · On-site · Ho Chi Minh City

- Designed **game shaders** in Shader Graph, ShaderLab, HLSL, and GLSL.
- Built **Unity Editor tools** for level design, game data, testing, and debugging.
- Developed games in **Unity, Unreal Engine 5, and Unreal Editor for Fortnite (UEFN)** with Verse.
- Ported existing mobile and PC projects to **WebGL** and extended their gameplay.

SKILLS

Shading	HLSL, GLSL, ShaderLab, Shader Graph, Raymarching / SDF
VFX & Rendering	Particle System, VFX Graph, Niagara, URP, Post-processing, Sprite Rigging
Art Tools	Blender, Maya, Photoshop, Substance Designer, Substance Painter
Engines	Unity, Unreal Engine 5, UEFN
Graphics APIs	DirectX 11, OpenGL
Optimization	Unity Profiler, Frame Debugger, RenderDoc, Draw-call Reduction, LOD, Texture Compression, WebGL Performance
Pipeline & Tools	Unity Editor Scripting, ScriptableObject, Object Pooling, Batch-mode CLI, Python Automation, AI-assisted Workflows, Git, Jenkins, Jira
Languages	C/C++/C#, Python, JavaScript

GAMES

Match-3 Game – Unity Engine

buciupi.github.io/match3-game-build

- Created in-game VFX that combine the Particle System, Animator, and sprite rigging.
- Built an effect manager that reuses effects across multiple object pools.

EDUCATION

Bachelor of Software Engineering · Ho Chi Minh City University of Technology (HUTECH) · 2018 – 2023 · GPA 3.26/4

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FEATURED PIPELINE

AI Game VFX: Effect Pipeline – Unity 6 (URP)

Personal project · 2026

An engine-independent VFX knowledge base and pipeline; each effect is described as data and built into engine assets.



- Created a generic Unity Editor builder that generates effect prefabs from JSON specs and checks performance budgets and art rules on every build.
- Built a VFX settings database (2,478 effects, 8,414 layers) and a 2,511-image texture library, with Python tools to query and audit them.
- Built a 120-effect set with the pipeline.
- Automated validation, builds, and tests through a Unity batch-mode CLI.
- Integrated AI image generation (Gemini) and AI coding agents (Claude Code) into the art and engineering workflow.

SHADERS & RENDERING

Unity Shaders & VFX

buciupi.github.io/assets/projects/unity-shader.html

- **Shaders:** shell texturing, hex shield, force field, hologram mapping, stylized foliage, toon water, pixelate and toon with shadow mapping
- **VFX:** VFX Graph particles, sword slash trail, healing spell, electric VFX, fractured sphere
- **After Cyanilux tutorials:** holofoil card, book flipping
- **Tools:** Shader Graph, ShaderLab, GLSL, HLSL

Unreal Engine 5 Custom Materials

buciupi.github.io/assets/projects/unreal-engine-shader.html

- **Shaders:** vertex displacement, texture packing, interior mapping, toon shading, tessellation, procedural hex grid
- **Post-processing:** blur, color filtering, lens masking, rain drops and drips, underwater with caustics and decals

DirectX 11 Render Pipeline

buciupi.github.io/assets/projects/directx11.html

- **Rendering:** forward and deferred, PBR lighting, shadow mapping, SSAO, compute-shader blur
- **HLSL:** vertex, compute, geometry, pixel, tessellation
- **Techniques:** terrain layers, water displacement, GPU instancing, occlusion culling, stencil and planar projection, billboards, texture packing

ARTIST TOOLS

ArtWorkflow

AI map art tool · Python

Large map → Tiles → ChatGPT → Best seam

Cuts a large map into overlapping tiles, refines each in ChatGPT, and rejoins them along the best seam, with versions and approvals.

Image to PSD

Layer rebuilder · Python

Flat image → RapidOCR → Type layers → PSD

Turns a flat design image into a layered Photoshop file with native type layers, using offline RapidOCR. Editor and CLI.

Asset Import Pipeline

Unity Editor · Batch mode

Art folder → Validate → Prefab → PR branch

Imports avatars and characters, builds prefabs and materials, and updates data and Addressables; batch mode pushes a review branch.

Asset Health Toolkit

Unity Editor · CLI

Scan → References → Duplicates → CLI

Reference Hunter, Duplicate Folder, and a GPU duplicate texture finder, with 10 CLI commands for scanning, repointing, and automation.

Rendering & Texture Tools

Unity Editor

Anim clips → Sample → Bounds

Mesh Bounds computes SkinnedMeshRenderer bounds from animation clips; Image Channel Packing packs up to 4 masks into a texture.

QA & Authoring Tools

Unity Editor

Routes → Validate → Heatmap

Card Preview, In-Game Avatar Changer, level design tools, and Walk Plan route generation with validation and a road-use heatmap.

Socket Automation & Rigging Guide

Python · Blender to Unity · EN & VI

SK_ + SM_ → Parent, align → Batch export

Automates socket parenting, alignment, renaming, and batch export; the guide covers hierarchy, LRA, skinning limits, and FBX export.

Graphics Studies

github.com/BuCiuPi/MeshProcedural

Raymarching · Mesh API · Burst · DOTS

Raymarched SDF scenes; procedural meshes with Unity's advanced Mesh API, Jobs, and Burst; a DOTS study with Entities and IJobEntity.